

HR-FONT: COMPLETING FONT FAMILIES ACROSS SCRIPTS

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ABSTRACT

We introduce HR-Font, a framework for extending a font family to new scripts from a few Chinese reference glyphs. The central challenge is to translate a family’s visual language into characters whose structures differ from those observed. HR-Font addresses this challenge through two complementary forms of conditioning. Mean-Delta expresses same-character examples from a font bank as a multi-scale variation prior relative to neutral content. Target-character appearance completion learns how reference appearance should be expressed in the requested glyph, using target-glyph appearance descriptors as training supervision. The resulting conditions guide a diffusion generator toward recognizable characters with a coherent family identity. We also formulate a cross-script completion benchmark that evaluates character identity, reference consistency, and family compatibility, with script-aware data curation and controlled reference budgets.

1 INTRODUCTION

A designer extending a Chinese typeface to Latin or kana needs more than a recognizable outline. The new glyphs should belong to the same family: weight, contrast, terminals, and decorative treatment should feel coherent across scripts. A few Chinese examples provide evidence of that visual language, but they leave its realization in the target characters open. The design of a Latin *g*, for example, requires choices that are not directly visible in a Chinese reference glyph. Cross-script font completion therefore calls for both a vocabulary of target-character variations and a model of how the observed appearance applies to each character.

Few-shot generators learn to synthesize glyphs from content and reference images. FontDiffuser uses conditional diffusion and multi-scale content aggregation (Yang et al., 2023); FSFont learns fine-grained correspondence between content and reference features (Tang et al., 2022). Font-bank methods such as CF-Font enrich content representations using corresponding glyphs from basis fonts (Wang et al., 2023). These approaches provide useful ingredients for completion. The cross-script setting makes a further distinction important: available glyph variations describe possible designs, whereas the reference images specify the family to which the result should belong.

We develop HR-Font around this distinction. The first component, *Mean-Delta*, represents a requested character through anchor-relative feature changes supported by a font bank. The neutral rendering supplies identity, and the residual supplies an adaptable variation prior. The second component, *target-character appearance completion* (TC), predicts an appearance descriptor for the requested character from its neutral content and the observed references. Training glyphs provide direct supervision for this prediction. Together, these components connect *how the character can vary* with *how the family’s appearance should be realized*. Figure 1 summarizes the model.

This formulation also shapes the evaluation. Matching a held-out target image is useful, but a plausible completion can differ from that particular design. We therefore assess three outcomes: whether the glyph has the requested identity, whether it expresses the references’ visible attributes, and whether it fits the target family across scripts. Script-aware curation establishes which font–character pairs provide meaningful supervision, while fixed reference episodes make comparisons reproducible.

Inputs	Conditions for the generator
Neutral target glyph	$x_c^0 \rightarrow E_c(x_c^0)$ Character identity
Same-character bank + reference similarity	$\Delta_c^s = \sum_j \alpha_j [E_c^s(B_{j,c}) - E_c^s(x_c^0)]$ Mean-Delta: anchor-relative variation
Reference appearance + target-content query	$\{\psi(r_i)\}, u_c \rightarrow H \rightarrow \hat{a}_{f,c}$ TC: predicted target-character appearance
Synthesis	$G'_j = G_j + W_o \hat{a}_{f,c}$ Content + Mean-Delta + completed global / local references $\rightarrow$ diffusion denoising $\rightarrow \hat{y}_{f,c}$

Figure 1: Two complementary conditions for cross-script completion. Mean-Delta summarizes bank-supported changes relative to the neutral target glyph. TC predicts target-glyph appearance from reference descriptors and a content query. A diffusion generator combines both with reference features. The target image supervises TC during training; generation uses the requested content and observed references.

Our contributions are:

- A cross-script font completion formulation and evaluation protocol that distinguish character identity, reference consistency, and family compatibility.
- Mean-Delta, an anchor-relative, multi-scale glyph variation prior that connects a neutral content representation to corresponding examples in a training font bank.
- Target-character appearance completion, a supervised predictor that translates reference appearance into a target-conditioned descriptor for diffusion synthesis.

2 RELATED WORK

Cross-script and few-shot font generation. FTransGAN studies few-shot transfer between languages with multi-level attention (Li et al., 2021). FontDiffuser generates glyphs by conditional denoising and uses multi-scale content aggregation and style contrastive learning (Yang et al., 2023). FSFont uses content queries to aggregate fine-grained local reference styles (Tang et al., 2022). HR-Font builds on conditional generation and local reference evidence, and introduces a supervised intermediate prediction: the appearance of the requested target glyph. This prediction connects reference observations to the target character before the generator renders the image.

Font-bank content representations. CF-Font projects content features into a space spanned by corresponding glyphs from basis fonts (Wang et al., 2023). Mean-Delta shares the use of same-character bank evidence. Its interface expresses that evidence as a residual from a fixed neutral anchor and consumes the residual through the structural conditioning path, alongside separately supplied neutral content. The shared anchor keeps the content input stable while the residual conditions its adaptation. Our evaluation tests this interface under shared bank access and training budgets.

Appearance transfer and completion. Feature moments provide a compact appearance representation in neural style transfer; AdaIN aligns content-feature statistics with those of a style image (Huang & Belongie, 2017). Reference attention retrieves useful features from observed glyphs. TC adds a predictive objective: estimate how those observations are expressed by a different character. The target-content query and target-glyph supervision give this objective a concrete meaning. Multi-scale feature statistics provide the appearance representation, and the generator receives the prediction as an additive update to its global reference tokens. Local reference features remain available for spatially detailed synthesis.

3 CROSS-SCRIPT FONT COMPLETION

Let c be a requested target character and x_c^0 its rendering in a neutral font. For a target family f , the observed reference set is $\mathcal{R}_f = \{(a_i, r_i)\}_{i=1}^n$, where a_i is a Chinese character and r_i its rendering in f . The task is to generate $\hat{y}_{f,c}$ with the identity of c and an appearance compatible with f . The evaluation font is held out from training. A bank $\mathcal{B}$ contains training-font renderings of the reference and target characters. During training, an episode excludes its own font from donor retrieval.

Three evaluation dimensions. *Identity* measures whether the output represents the requested character. *Reference consistency* measures agreement with visible attributes such as weight, contrast, stroke endings, and texture. *Family compatibility* measures whether the generated glyph is a coherent extension of the family across scripts. The latter includes character-specific decisions whose exact realization is not shown in the reference set. Paired target-image measurements complement these dimensions with reconstruction diagnostics.

Script-aware supervision. A font file can contain several scripts with different design conventions, fallback glyphs, or inconsistent coverage. We curate valid font-character pairs and eligible donors by script, preserving the font-level train/validation/test split. A mismatch in design conventions calls for family-level inspection; automatic similarity alone does not determine validity. The released protocol records pair validity, donor eligibility, and the reference episode used for each output.

4 METHOD

4.1 OVERVIEW

We use a conditional diffusion generator with frozen content and style encoders, E_c and E_s . The content encoder supplies multi-scale features of the neutral target. The style encoder supplies global reference tokens and, for the local-reference configuration, local reference features. Mean-Delta conditions the structural interaction path. TC updates the global reference tokens with predicted target-character appearance. The denoiser thus receives

$$\hat{\epsilon} = \epsilon_\theta(x_t, t, E_c(x_c^0), \Delta_c(\mathcal{R}_f), G'(\mathcal{R}_f, c), L(\mathcal{R}_f)). \quad (1)$$

Here x_t is the noisy target image, G' denotes the appearance-completed global tokens, and L denotes local reference tokens when present. We evaluate both global and global-plus-local configurations using the same TC definition.

4.2 MEAN-DELTA: A GLYPH VARIATION PRIOR

The bank supplies examples of how the requested character is realized in different fonts. To prioritize a relevant neighborhood, we compare the target references with bank renderings of the same reference characters:

$$s_j(\mathcal{R}_f) = \frac{1}{n} \sum_{i=1}^n \cos(\bar{E}_s(r_i), \bar{E}_s(B_{j,a_i})), \quad (2)$$

$$\alpha_j = \frac{\exp(s_j/\tau)}{\sum_{\ell \in \mathcal{J}_c(\mathcal{R}_f)} \exp(s_\ell/\tau)}, \quad j \in \mathcal{J}_c(\mathcal{R}_f). \quad (3)$$

$\bar{E}_s$ is a pooled descriptor. $\mathcal{J}_c$ contains the top- K eligible training fonts after removing the episode font and script-invalid donors. The implementation uses $K = 10$ and $\tau = 0.07$.

For each content scale s , Mean-Delta is

$$\Delta_c^s = \sum_{j \in \mathcal{J}_c} \alpha_j [E_c^s(B_{j,c}) - E_c^s(x_c^0)]. \quad (4)$$

Every donor renders the same requested character. Subtracting a common anchor expresses the bank evidence in coordinates relative to the content already available to the generator. The resulting multi-scale tensors feed the deformable structural interaction modules.

Retrieval defines a reference-conditioned prior over available variations. Similarity in the observed script is a useful ranking signal; the generator learns how the resulting prior applies to the target script. The donor residual is therefore an adaptable condition rather than a prescribed target design. Uniform-weight and absolute-feature controls test the contribution of neighborhood weighting and the anchor-relative interface.

4.3 TARGET-CHARACTER APPEARANCE COMPLETION

TC learns the appearance of a target character from reference observations. Its input pairs a descriptor of each reference with a query describing the requested content. Its teacher is the appearance descriptor of the corresponding training target.

Appearance representation. Let $\phi_\ell(x)$ denote three frozen VGG16 (Simonyan & Zisserman, 2015) feature maps with channel counts 64, 128, and 256. We summarize each map by its spatial channel means and population standard deviations:

$$\psi(x) = \text{concat}_{\ell=1}^3 [\mu_{HW}(\phi_\ell(x)), \sigma_{HW}(\phi_\ell(x))] \in \mathbb{R}^{896}. \quad (5)$$

Training-target statistics define a per-dimension standardization S shared by reference descriptors and target teachers. The descriptors summarize multi-scale activation strength and variability. They give TC a compact appearance target while the diffusion generator handles the spatial arrangement of strokes.

Content-conditioned prediction. The five pooled content scales form $u_c \in \mathbb{R}^{707}$. Linear projections map u_c and the standardized reference descriptors into a 256-dimensional attention space:

$$q_c = W_q u_c, \quad v_i = W_r S(\psi(r_i)), \quad (6)$$

$$h_c = \text{LN}(q_c + \text{MHA}(q_c, \{v_i\}, \{v_i\})), \quad (7)$$

$$\hat{a}_{f,c} = \text{MLP}([h_c; q_c]) \in \mathbb{R}^{896}. \quad (8)$$

The attention uses eight heads and masks padded references. The content query also enters the prediction MLP directly, allowing character-dependent predictions even for a single reference. The same family can consequently yield different descriptors for different requested characters.

Supervision and injection. For training target $y_{f,c}$, the teacher is $a_{f,c}^* = S(\psi(y_{f,c}))$. TC minimizes

$$\mathcal{L}_{\text{comp}} = \text{SmoothL1}(\hat{a}_{f,c}, \text{stopgrad}(a_{f,c}^*)). \quad (9)$$

A learned projection $W_o : \mathbb{R}^{896} \rightarrow \mathbb{R}^{1024}$ converts the prediction into a shared residual for the existing nine global tokens:

$$G'_j = G_j + W_o \hat{a}_{f,c}, \quad j = 1, \dots, 9. \quad (10)$$

W_o includes a bias; its weights and bias start at zero. This initialization begins joint optimization from the parent generator. The global tokens retain their existing spatial information, and local reference tokens supply additional detail. TC is trained on reference descriptors and neutral content, so it learns a target-appearance mapping complementary to the bank-derived variation condition.

4.4 TRAINING AND GENERATION

Training first fits the appearance predictor using cached descriptors. Joint training then optimizes the predictor, its output projection, and the denoiser:

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + 0.01\mathcal{L}_{\text{perc}} + 0.5\mathcal{L}_{\text{off}} + 0.01\mathcal{L}_{\text{comp}}. \quad (11)$$

$\mathcal{L}_{\text{diff}}$ is the epsilon-prediction mean squared error; the perceptual and offset terms follow the inherited generation backbone. The content, style, and VGG encoders stay frozen. Each training episode uses one to eight Chinese references and a valid target glyph from the same font.

The TC teacher is used only for its supervised loss. The denoiser always receives the predicted descriptor, including during training. Classifier-free condition dropout masks the TC residual together with the reference condition, while the completion loss uses the original observed inputs. The structural condition follows the shared source-drop policy. At inference, TC is evaluated once for each content-reference pair and reused across denoising steps.

Table 1: Main comparison under the cross-script protocol. Identity and style preference are percentages; family compatibility uses the calibrated evaluator. Empty cells await completed evaluation.

Method	Identity $\uparrow$	Style pref. $\uparrow$	Family $\uparrow$	LPIPS $\downarrow$
FontDiffuser	—	—	—	—
FTransGAN	—	—	—	—
CF-Font	—	—	—	—
FSFont	—	—	—	—
Mean-Delta baseline	—	—	—	—
HR-Font	—	—	—	—

5 EXPERIMENTS

5.1 DATA AND SETTINGS

The corpus contains 260 fonts partitioned into 228 training, 16 validation, and 16 test fonts. The inventory comprises 295 target characters and 338 Chinese reference characters. Protocol A renders RGB glyphs at 96×96 , with a shared font size within each font and neutral Noto Sans CJK Regular content. The current clean training mapping retains 56,429 target pairs. The stratified evaluation panel spans 47 characters covering Latin, diacritics, kana, and phonetic symbols; validity filtering retains 704 test pairs.

Reference-budget evaluation uses nested sets of 1, 2, 4, and 8 characters. All methods in a comparison receive the same ordered reference set, and Mean-Delta retrieval uses that episode’s references. The main reference manifest and checkpoint selection are finalized on validation data. The appendix records reference characters, bank eligibility, sampler, and provenance.

Training settings. The current joint experiments use eight V100 GPUs, eight samples per device, FP16 arithmetic, and gradient clipping at 1.0. The denoiser learning rate is 10^{-5} ; new TC parameters use 10^{-4} . Short joint stages run for 5,000 updates with 200 warmup updates and checkpoints at 2,500 and 5,000. TC pretraining uses batch size 256, learning rate 3×10^{-4} , and up to 2,000 updates. Matched continuation controls start from the same parent weights with the same training schedule. Complete resolved settings appear in Appendix A.

5.2 COMPARISONS AND MEASUREMENTS

External comparisons cover conditional diffusion (FontDiffuser), cross-language transfer (FTransGAN), font-bank content fusion (CF-Font), and fine-grained reference attention (FSFont). Each method is accompanied by its reference budget, bank access, pretraining data, and cross-script adaptation. Internal comparisons hold the content/style encoders and reference interface fixed while changing the structural source or TC. The strongest validation baseline is retained in the main comparison.

We report character recognition accuracy, blind paired style preference, and family compatibility. Human comparison displays the neutral target and the same reference set with randomized method positions and an explicit tie option. The learned compatibility evaluator is trained and calibrated separately from the generator; its definition, calibration, and agreement with human judgments are reported with the final results. L1, SSIM, and LPIPS provide paired reconstruction diagnostics. Results are aggregated by font and script over the same valid examples.

5.3 CONTRIBUTION STUDIES

Glyph variation prior. Matched structural-source experiments compare the reference-derived structural input with Mean-Delta while preserving the reference interface. A same-bank absolute-feature control tests anchor-relative conditioning. A uniform-weight control measures the contribution of reference-based donor ranking. Each comparison uses the same training budget and reports identity and appearance jointly.

Table 2: Contribution studies. Each row identifies one experiment and its outcome; the execution plan stages these studies after the paired TC continuations.

Question	Matched comparison	Result
Structural source	Mean-Delta versus reference-derived structure	–
Residual interface	Anchor-relative versus absolute bank features	–
Retrieval weighting	Similarity-weighted versus uniform donors	–
TC increment	TC versus parent continuation, on each interface	–
Character conditioning	Content query versus constant query	–
Appearance supervision	Supervised TC versus capacity-matched branch	–
Complementarity	Mean-Delta $\times$ TC training study	–

Case	References	Baseline	Mean-Delta	HR-Font	Target
Stroke style					
Decoration					
Difficult completion					

Figure 2: Qualitative comparison layout. Rows cover stroke style, decorative treatment, and difficult cross-script completion. Blank cells will contain matched generated glyphs and the corresponding target design.

Target-character appearance completion. We compare TC with ordinary continuation from each of the G2 and G2-RL parents. These pairs test the incremental value of appearance completion on global and global-plus-local reference interfaces. We also evaluate a reference-only predictor with a constant content query and a same-capacity branch trained without target-descriptor supervision. The reference-only predictor tests character conditioning. The latter control measures the value of the complete supervised initialization and training recipe.

Complementarity. A factorial experiment varies Mean-Delta and TC under a fixed reference interface. The comparison tests their separate and combined effects on family compatibility and reference consistency. The initialization and structural-source treatment are shared within this study. This is a training comparison; inference-time condition removal is reported separately as sensitivity analysis.

5.4 REFERENCE BUDGETS AND QUALITATIVE ANALYSIS

Reference-budget curves evaluate the same checkpoints at 1, 2, 4, and 8 shots. They measure how additional observations affect identity, appearance, and family coherence. Qualitative comparisons include ordinary stroke styles, decorative fonts, and difficult cross-script realizations. For each selected row, all methods share the target, references, sampler, and noise. Figure 2 reserves the comparison layout for the completed outputs.

6 DISCUSSION

The two conditions provide different forms of information. Mean-Delta brings target-character examples into the generator’s content coordinates. TC learns a target-specific appearance prediction from observed references. Their combination gives the generator a structural prior and a supervised appearance destination. The contribution studies measure how these roles translate into image quality.

The current representation operates on 96×96 raster glyphs. Spatial feature statistics summarize appearance compactly but do not specify the location of every ornament or terminal. Fine decorative placement remains a synthesis problem for the denoiser and its local reference interface. Bank

coverage also determines the available examples for rare target designs. Extending the formulation to higher-resolution glyphs and complete typographic layout is a natural next step.

7 CONCLUSION

HR-Font frames cross-script font generation as coherent family completion. Mean-Delta provides a bank-supported variation prior relative to neutral content, and target-character appearance completion predicts how reference observations apply to the requested glyph. The accompanying evaluation separates identity, reference consistency, and family compatibility. Together, the formulation and two conditioning components establish a practical framework for extending a font’s visual language across scripts.

REPRODUCIBILITY STATEMENT

The supplement records data mappings, cache fingerprints, reference episodes, parent checkpoints, resolved training configurations, and evaluation scripts. The paper’s editable figure sources accompany the draft. Final results will include measured training and inference costs.

ETHICS STATEMENT

Font completion can reduce repetitive design work and improve script coverage. Font licenses and source provenance must govern research use and redistribution. The release will document permitted assets and the consent, compensation, and review procedures for human evaluation.

AI USE STATEMENT

AI tools assisted with research discussion, code review, drafting, and figure layout. Experimental measurements come from the recorded training and evaluation runs. The authors review the resulting text and artifacts and take responsibility for the submission.

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A REPRODUCIBILITY DETAILS

Table 3: Current implementation settings. Final run hashes and measured costs are filled from the selected artifacts.

Setting	Value
Image resolution	96×96 RGB, Protocol A
Font split	228 / 16 / 16
Valid training target pairs	56,429
Mean-Delta retrieval	top-10, temperature 0.07
TC reference descriptor	VGG16 means and population stds, 896 dimensions
TC content descriptor	five pooled content scales, 707 dimensions
TC predictor	width 256, eight-head attention, MLP
TC output	shared 1024-dimensional residual on nine global tokens
Optimizer	AdamW, (0.9, 0.999), $\epsilon = 10^{-8}$, weight decay 0.01
Joint batch / precision	$8 \times 8 = 64$ / FP16
Joint learning rates	denoiser 10^{-5} ; TC 10^{-4}
Schedule	constant after 200 warmup updates; 5,000 updates
Gradient clipping	1.0
Loss coefficients	perceptual 0.01; offset 0.5; completion 0.01
Reference / source dropout	0.10 / 0.25
Parent checkpoint hashes	—
Mapping and cache hashes	—
Training GPU hours / peak memory	—
Inference latency	—

Implementation details. VGG descriptors use ImageNet normalization and FP32 statistics. Only clean training targets fit descriptor standardization. The predictor is pretrained for at most 2,000 updates with 100 warmup updates. Joint stages load parent weights and create a new optimizer and scheduler. The local-reference learning rate is preserved within each matched pair. The default joint configuration uses the denoiser rate for existing local projections. The separate local-reference pilot uses 10^{-4} for its new local projection.

Randomness and uncertainty. Training uses random state 3407. Evaluation pairs reference episodes and diffusion noise across methods. Planned 95% intervals resample fonts while keeping paired method outputs together. These intervals describe variation across evaluated fonts.

Reference episodes and sampler. The primary nested episode is specified by Unicode code points: U+5C71, U+6C34, U+4E91, U+6708, U+91D1, U+77F3, U+5FC3, U+6587. The 1/2/4/8-shot sets use prefixes of this list. Earlier exploratory boards used a different reference episode and retain separate provenance. The intended common inference protocol is DPM-Solver++ order 2, 20 steps, and guidance scale 7.5. The TC sampling path is validated against the common protocol before final paired reporting.

Model selection. Validation panels guide model and checkpoint selection using appearance, readability, and identity. The final selection uses the full clean validation split. Exploratory test visualizations are retained with their dates; subsequent selection uses validation data. All reported methods are scored over an explicitly intersected valid sample set.

B EVALUATION COMPLETION

The learned evaluator is currently being trained. Its architecture, training data, disjoint calibration split, score definition, and human-agreement results will be inserted here when finalized. The human study will specify sample size, rating instructions, ties, exclusions, consent, and font-clustered uncertainty. The main tables intentionally reserve numerical results until the matched runs and this evaluation are complete.

C ADDITIONAL RESULTS

Table 4: Reference-budget evaluation of the selected HR-Font checkpoint.

References	Identity $\uparrow$	Style pref. $\uparrow$	Family $\uparrow$	LPIPS $\downarrow$
1	—	—	—	—
2	—	—	—	—
4	—	—	—	—
8	—	—	—	—